Sungmin Kim, Ph.D. Candidate

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Research interests:

- Efficient pattern matching algorithms under various equivalence relations
- Subsequences and Simon's congruence
- Automata and formal languages

Education

2022 – Present **Ph.D. candidate in Computer Science, Yonsei University**
Ph.D. Supervisor: Prof. Yo-Sub Han
Expected graduation: August 2026

2017 – 2022 **BA in English Language and Literature, Yonsei University**
BSE in Computer Science, Yonsei University
Double major, Highest honors at graduation.

Research Publications

Journal Articles

1. P. Fleischmann, **S. Kim**, T. Koß, *et al.*, "Matching patterns with variables under Simon's congruence," 2026.
2. Y.-S. Han, **S. Kim**, S.-K. Ko, and K. Salomaa, "Existential and universal width of alternating finite automata," *Information and Computation*, p. 105 337, 2025, invited paper.
3. **S. Kim** and Y.-S. Han, "Approximate Cartesian tree pattern matching," *Theoretical Computer Science*, p. 115 506, 2025.
4. **S. Kim**, S.-K. Ko, and Y.-S. Han, "Simon's congruence pattern matching," *Theoretical Computer Science*, vol. 994, p. 114 478, 2024.
5. **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On Simon's congruence closure of a string," *Theoretical Computer Science*, vol. 972, p. 114 078, 2023, invited paper.

Conference Proceedings

1. **S. Kim**, H. Hong, T. Lim, Y.-S. Han, and K. Salomaa, "Decomposing regular languages under shuffle along trajectories," in *Proceedings of the 30th International Conference on Implementation and Application of Automata (CIAA)*, to appear, 2026.
2. **S. Kim**, H. Jin, and Y.-S. Han, "Pattern matching under $\mathcal{R}$-congruence," in *Proceedings of the 30th International Conference on Implementation and Application of Automata (CIAA)*, to appear, 2026.
3. **S. Kim** and Y.-S. Han, "Pattern mining under Simon's congruence," in *Proceedings of the 29th International Conference on Developments in Language Theory (DLT)*, vol. 16036, 2025, pp. 182–196.
4. **S. Kim** and Y.-S. Han, "Approximate Cartesian tree pattern matching," in *Proceedings of the 28th International Conference on Developments in Language Theory (DLT)*, vol. 14791, 2024, pp. 189–202.
5. P. Fleischmann, **S. Kim**, T. Koß, *et al.*, "Matching patterns with variables under Simon's congruence," in *Proceedings of the 17th International Conference on Reachability Problems (RP)*, vol. 14235, 2023, pp. 155–170.

6. Y.-S. Han, **S. Kim**, S.-K. Ko, and K. Salomaa, "Existential and universal width of alternating finite automata," in *Proceedings of the 25th International Conference on Descriptional Complexity of Formal Systems (DCFS)*, vol. 13918, 2023, pp. 51–64.
7. **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On the Simon's congruence neighborhood of languages," in *Proceedings of the 27th International Conference on Developments in Language Theory (DLT)*, vol. 13911, 2023, pp. 168–181.
8. **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On Simon's congruence closure of a string," in *Proceedings of the 24th International Conference on Descriptional Complexity of Formal Systems (DCFS)*, vol. 13439, 2022, pp. 127–141.
9. **S. Kim**, S.-K. Ko, and Y.-S. Han, "Simon's congruence pattern matching," in *Proceedings of the 33rd International Symposium on Algorithms and Computation (ISAAC)*, vol. 248, 2022, 60:1–60:17.

Under Review

1. **S. Kim** and Y.-S. Han, "Pattern mining under Simon's congruence," invited and submitted to special issue of Information and Computation.
2. **S. Kim**, Y.-S. Han, S.-K. Ko, and K. Salomaa, "On the Simon's congruence neighborhood of languages," submitted to Theoretical Computer Science.

Talks and Seminars

Conferences	Presented at DLT'25, DLT'24, DCFS'23, DLT'23, DCFS'22 (online), ISAAC'22
Workshops	Presented at WAAC'24, FWAC'23
Fall 2024	Orthogonal range searching, Yonsei CS Theory Student Group
	3SUM-hard problems in computational geometry, Yonsei CS Theory Student Group
Summer 2024	Cartesian tree edit distance, Yonsei CS Theory Student Group
Spring 2024	The rate distortion theory, Yonsei CS Theory Student Group
	Information theory in practice: the JPEG algorithm, Yonsei CS Theory Student Group
Winter 2024	Basics of Simon's congruence, Yonsei CS Theory Student Group
Fall 2023	Shor's algorithm, Yonsei CS Theory Student Group
Summer 2023	Invited talk: Alternating finite automata, University of Göttingen
Spring 2023	Optimal mechanism design for the seller's auction design problem, Yonsei CS Theory Student Group

Academic Activities and Services

Reviewer	Conferences: DLT'26 / CIAA'26 / DCFS'26 / STACS'26 / CIAA'25 / DLT'25 / DCFS'25 / ISAAC'24 / DLT'24 / CIAA'24
	Journals: Computer Science Review / Journal of Automata, Languages and Combinatorics / Theoretical Comptuer Science / International Journal of Foundations of Computer Science
2024 – today	Co-organized Yonsei CS Theory Student Group Seminars
2023 – 2025	Teaching Assistant: Data Structures / Automata and Formal Languages / Theory of Computation / Analysis of Algorithms
2022 – 2025	Problem author, ICPC regional (Asia, Seoul)

Experience

2025 – 2026	Participation in Industry–University Collaborative Project, LG
2024 – 2025	Lab manager for the Theory of Computation Lab (Yonsei University)
Summer 2023	Visiting research at Göttingen University, Germany (2 weeks) Topic: Matching patterns with variables under Simon's congruence
Summer 2022	Visiting research at Queen's University, Canada (2 weeks) Topic: Alternating finite automata, Simon's congruence neighborhood

Miscellaneous

Other Skills

Languages	Korean (native speaker), English (fluent), Chinese (medium), Japanese (beginner), German (beginner)
Programming	C++, Java, Python

Scholarships

2022 – 2024	**Accelerated degree program scholarship**
2023	**Academic research support ARF fellowship grant**
2017–2021	**Performance and merit scholarships during BA/BSE studies**

As of: May , 2026